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SANDIPAN-DEY

main.py ×

```python
        handler.write(img_data)
    return img

def generate_earth_image(angle):
    prompt = f"Realistic Earth from space at {angle} degrees angle"
    image_url = generate_base_image(prompt)
    image_filename = f"earth_{angle}_degrees"
    image = download_image(image_url, image_filename)
    return image

#angles = range(0, 360, 10)
#earth_images = [generate_earth_image(angle) for angle in angles]

angles = range(0, 360, 10)
image_filenames = [f"earth_{angle}_degrees.jpg" for angle in angles]

resized_earth_images = []
for filename in image_filenames:
    img = Image.open(filename)
    resized_img = img.resize((256, 256), Image.ANTIALIAS)
    resized_earth_images.append(resized_img)

output_gif = "rotating_earth.gif"
resized_earth_images[0].save(
    output_gif,
    save_all=True,
    append_images=resized_earth_images[1:],
    duration=100,
    loop=0,
    optimize = True
)
```

99% (55:20)

Guide ×

Collapse

Optimizing Animation Settings

We will explore various options to optimize the GIF animation created using the resized Earth images. We will focus on adjusting the frame duration, loop count, and image optimization settings to make the animation smoother and more visually appealing.

Adjusting the duration

You can adjust the duration of each frame in the animation to make the rotation appear faster or slower. The duration parameter in the save method determines the time each frame is displayed in milliseconds. A lower duration will result in a faster rotation, while a higher duration will make the rotation slower. Experiment with different values to find the optimal speed for your animation. Change the duration from 100 to 50 to 200 and compare.

```python
duration = 50  # Adjust the frame duration to your preference
```

TRY IT

Command was successfully executed.

Click here to refresh your earth gif

and

```python
duration = 200 , # Adjust the frame duration to your preference
```

TRY IT

Command was successfully executed.

Click here to refresh your earth gif

Adjusting Loop Count

You can control the number of times the animation loops by setting the `loop` parameter in the `save` method. A value of `0` indicates an infinite loop, while a higher value specifies the exact number of loops. For example, if you want the animation to loop only three times, set the loop parameter to `3`

```python
loop_count = 3  # Adjust the loop count to your preference
```

TRY IT

Command was successfully executed.

Click here to refresh your earth gif

Image Optimization

By default, the PIL library doesn't optimize the images in the animation. You can enable image optimization by setting the `optimize` parameter to `True` in the `save` method. This can reduce the file size of the animation while maintaining visual quality.

```python
optimize = True  # Enable image optimization
```

TRY IT

Command was successfully executed.

Click here to refresh your optimized earth gif

Now, let's put everything together and create an optimized GIF animation:

```python
output_gif = "optimized_rotating_earth.gif"
resized_earth_images[0].save(
    output_gif,
    save_all=True,
    append_images=resized_earth_images[1:],
    duration=duration,
    loop=loop_count,
    optimize=optimize
)
```

TRY IT

Command was successfully executed.

Click here to refresh your eartg gif
With these optimizations in place, you should now have a smoother, more visually appealing animation that showcases the Earth rotating using the images generated by the DALL-E 2 API.

True or False:
Setting the optimize parameter to `True` in the `save()` method can help reduce the file size of the animation while maintaining visual quality.

- (●) True
- ( ) False

The optimize parameter, when set to True, enables image optimization in the save() method. This can help reduce the file size of the animation without significantly affecting the visual quality, resulting in a more efficient and visually appealing animation.

Check It!

Next